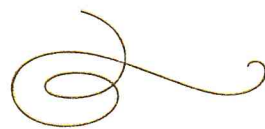
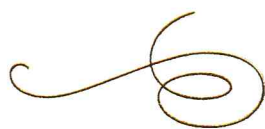


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(12) **United States Patent**
Yasunaga(10) **Patent No.: US 10,919,022 B2**(45) **Date of Patent: Feb. 16, 2021**(54) **EXHAUST GAS PURIFYING AGENT FOR
AUTOMOBILES AND METHOD OF
PRODUCING THEREOF**(71) Applicant: **EISHIN CO., LTD.**, Minato-ku (JP)(72) Inventor: **Chie Yasunaga**, Minato-ku (JP)(73) Assignee: **EISHIN CO., LTD.**, Minato-ku (JP)(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.(21) Appl. No.: **16/605,139**(22) PCT Filed: **Aug. 24, 2017**(86) PCT No.: **PCT/JP2017/030361**

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35/0033; B01J 35/026; B01J 37/0219;
B01D 53/94; B01D 2255/65; B01D

2255/707

USPC 502/404

See application file for complete search history.

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Maier & Neustadt, L.L.P.

(57)

ABSTRACT

The problem of the present invention is to provide an exhaust gas purifying agent for automobiles which is capable of improving the automobile fuel mileage as well as purifying the exhaust gas of the automobiles, and further extending the effective duration of the exhaust gas purifying agent for automobiles by a simple method of spraying the exhaust gas purifying agent for automobiles using only natural ingredients on the air filter.

The problem of the present invention can be solved by using a liquid containing a) tourmaline fine powder, b) porous material fine powder, c) fucoidan extracted from seaweeds, d) amino peptides and/or alginic acids extracted from seaweeds in water as the exhaust gas purifying agent for automobiles by spraying on the air filter of automobiles.

5 Claims, 5 Drawing Sheets